CHAINSILIENCE AI: An Agentic, Digital-Twin Framework for Supply-Chain Disruption Intelligence

Ka Woon Ho, Ke Xin Cao, Gary Leng, Mo Xia, Andrew Li

ansonho31@gmail.com
caokexin5632@gmail.com
Lengg0701@gmail.com
mark.xia.mo@gmail.com
andrew.li3056@gmail.com

International Data Science Olympiad (IDSOL) 2026

AI-powered resilience for SMEs, startups, and semiconductor supply networks

Abstract — Chainsilience AI is a proposed disruption-intelligence and response platform for small and medium-sized enterprises (SMEs), startups, and semiconductor-related businesses. It converts fragmented economic, geopolitical, environmental, regulatory, logistics, and industry reporting into company-specific operational decisions. The pipeline collects external information, extracts and verifies events with AI agents, maps them to a supply-chain knowledge graph and digital twin, estimates exposure and operational impact, and produces explainable recommendations, draft communications, dashboard updates, and alerts. Unlike generic news monitoring, the system combines public evidence with private enterprise data such as suppliers, sites, routes, inventory, production schedules, and customer orders. This paper defines the architecture, risk model, governance controls, evaluation protocol, and a retrospective Taiwan-earthquake scenario. Chainsilience AI is currently implemented as a functional prototype demonstrating news relevance filtering, disruption extraction, company-specific exposure analysis, mitigation recommendation, and dashboard presentation. Advanced enterprise integrations, large-scale evaluation, and controlled external actions remain planned capabilities.

***Keywords*- supply-chain resilience, agentic AI, digital twin, knowledge graph, semiconductor risk, retrieval-augmented generation, LLM reasoning, SME, model context protocols, natural language processing**

I. INTRODUCTION AND PROBLEM

Modern supply chains are exposed to interacting shocks rather than isolated failures. In 2024, disruption around the Red Sea, Suez Canal, and Panama Canal lengthened routes, increased freight-rate volatility, and raised costs [1]. Semiconductor ecosystems add concentrated materials, specialized facilities, long qualification cycles, and geopolitical controls; a U.S. industrial-base assessment identifies fragile supply chains, customer concentration, geopolitical factors, and limited transparency as material risks [2].

Large enterprises may purchase specialist intelligence and maintain dedicated risk teams. Many SMEs instead depend on supplier emails, spreadsheets, search alerts, and periodic meetings. The decisive information may exist, but it is separated from the operational context required to answer: Which order is threatened? How long will stock last? Which customer commitment fails first? What is the least-cost mitigation strategy?

A. Problem Statement

Manual monitoring creates three delays. First, analysts must discover a relevant event among high-volume reporting. Second, they must verify location, affected entities, products, timing, and severity. Third, they must translate the event into dependencies hidden across business systems. Conventional disruption models support preparedness and recovery, but their value depends on timely, current inputs [3]. Chainsilience AI addresses this event-to-decision gap.

B. Research Gap

Digital supply-chain twins can represent the network state and test disruption scenarios [4], while ISO 23247 provides general digital-twin principles for manufacturing [5]. Knowledge graphs can reveal multi-tier dependencies between entities such as suppliers and shipping routes and support risk reasoning [6], and recent work shows that large language models (LLMs) can extract supply-chain related news from public sources [7]. The remaining design challenge is to synthesize these capabilities into a governed operational workflow that preserves evidence, quantifies uncertainty, and requires authorization before consequential actions.

C. Objectives

The project has five objectives: (1) detect and normalize relevant external events; (2) connect events to enterprise assets and commitments; (3) estimate revenue at risk, delay, inventory, cost of mitigation, and production stoppage probability; (4) generate traceable mitigation options and communications; and (5) evaluate timeliness, accuracy, usefulness, and safety.

D. Scope and Status

Implemented/prototyped: intended for demonstrated modules and interfaces only after they exist. **Planned:** architecture, integrations, scoring, simulations, and controlled agent actions described here. **Assumed:** enterprise data availability, access rights, and scenario inputs. **Illustrative:** all case-study probabilities, costs, and recommended allocations. This explicit labeling prevents a design proposal from being mistaken for production evidence.

Question	System output
What happened?	Verified event + source provenance
Why relevant?	Mapped suppliers, parts, sites, routes
What follows?	Delay, days of cover, order impact
What now?	Ranked actions, cost, trade-offs, owner

TABLE I. FROM EXTERNAL SIGNAL TO OPERATIONAL DECISION

II. CHAINSILIENCE AI SOLUTION AND CONTRIBUTION

A. Value Proposition

Chainsilience AI is a continuously refreshed decision-support layer above a firm's existing systems. It does not replace procurement, enterprise resource planning (ERP), manufacturing execution, transport management, or human judgment. It combines public evidence with authorized internal data and presents the smallest set of decisions needed to protect service, margin, and continuity.

The primary users are procurement managers, operations planners, startup founders, logistics teams, finance controllers, and risk owners. A user can move from an alert to the supporting articles, extracted facts, affected graph path, assumptions, simulation result, recommended action, and approval history. This evidence chain is essential because the same event may be severe for one company and irrelevant for another, depending on their respective suppliers, routes, ports, etc.

B. Differentiators

Capability	Generic monitor	Chainsilience AI
Signal	Keyword/news alert	LLM relevance filtering and cross-source verification
Context	Sector or geography	Company graph + digital twin
Impact	Narrative severity	Orders, delay, revenue at risk, inventory
Response	Analyst interpretation	Simulation of multi-versioned response options
Audit	Link or summary	Claim-level provenance + approvals

TABLE II. DIFFERENTIATION FROM GENERIC MONITORING

The contribution is not a claim that an LLM can autonomously manage a supply chain. It is a governed composition of retrieval, structured extraction, deterministic calculations, graph traversal, simulation, optimization, and human authorization. LLMs handle language-intensive tasks; calculators and solvers handle quantities; policy engines determine which actions are permitted.

C. Stakeholder Outcomes

For SMEs, the intended outcome is earlier, more consistent analysis without a large intelligence team. For semiconductor firms, the graph can represent foundries, outsourced assembly and test, materials, equipment, utilities, routes, and qualification constraints. For customers and suppliers, templated communications reduce response time while preserving approval.

D. End-to-end Workflow

Execution begins when an authorized company administrator defines the permitted data boundary and uploads or connects the company's product records, bills of materials, component mappings, suppliers and manufacturing sites, approved alternatives, inventory and in-transit stock, purchase and customer orders, production schedules, logistics routes, lead times, costs, and service priorities.

The platform validates and standardizes these records, filters out contradicting or faulty information after review, and creates a versioned supply-chain knowledge graph and digital-twin baseline. Chainsilience AI then monitors authorized external sources, verifies potential disruptions, identifies affected dependency paths, and simulates the operational and financial consequences of taking no action.

The LLM proposes evidence-backed candidate mitigation actions, such as reallocating inventory, rerouting, adjusting production schedules, or transferring orders to qualified alternative suppliers. Deterministic and optimization services compare the candidates using impact metrics such as reduced risk, recovery time, and implementation cost.

A designated decision-maker reviews the supporting sources, affected supply-chain relationships, assumptions, calculations, and alternative strategies. The decision-maker may revise assumptions, modify the proposed response, rerun the simulation, and approve the exact mitigation strategy and any related communications.

Once approved, Chainsilience AI initiates only the actions permitted by company policy, such as creating procurement tasks, preparing supplier emails, notifying internal teams, or passing authorized changes to connected enterprise systems. The dashboard then displays the original no-action risk, projected post-mitigation risk, implementation progress, and realized operational impact as supplier confirmations, inventory updates, shipment information, and production data become available.

Projected risk reduction is clearly distinguished from realized risk reduction until supporting operational evidence has been received.

III. SYSTEM ARCHITECTURE AND DATA PIPELINE

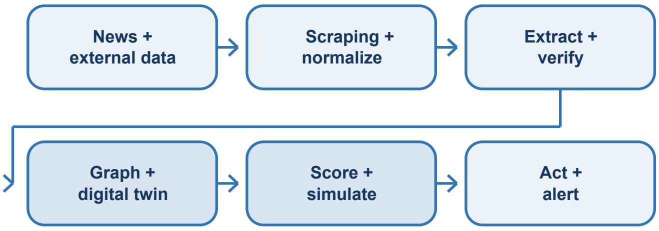

Fig. 1. Core Chainsilience AI pipeline. Each transition stores confidence, provenance, timestamp, and version.

A. News Collection/Scraping and Normalization

Connectors ingest permitted sources: government advisories, earthquake and weather feeds, trade and customs notices, port and carrier updates, company disclosures, trusted media, and licensed datasets. Documents are hashed, timestamped, language-detected, deduplicated, and retained with source metadata. Internal connectors read only the fields required from supplier master data, bills of material, inventory, orders, routes, and schedules.

B. AI-Agent Extraction and Verification

An LLM based extraction AI agent determines whether each collected news item is relevant to the company's suppliers, facilities, routes, products, or components. It converts unstructured text into a schema containing event type, entities, locations, products, start time, expected duration, severity, quantities, and cited evidence spans. A verifier agent filters out potentially unreliable and unsupported information after further review by comparing independent sources, detecting contradictions, checking temporal and geographic consistency, and assigning confidence levels. Retrieval-augmented generation (RAG) grounds summaries in retrieved records [9]; provenance follows the entity-activity-agent concepts standardized by W3C's PROV-O [10].

Agentic workflows may interleave reasoning with tool use, following patterns demonstrated by ReAct [8]. Chainsilience AI constrains this behavior: tools expose typed inputs, least-privilege permissions, timeouts, and auditable results. The LLM cannot directly alter enterprise records or send external communications unless a workflow policy and designated human approver permit it.

C. Enterprise Context Layer

The event object is joined to enterprise identifiers. Entity resolution uses names, addresses, geospatial distance, corporate ownership, product codes, and user-confirmed aliases. Unresolved matches enter a review queue rather than silently becoming graph facts. Confidence therefore travels with each edge and each downstream conclusion.

Layer	Representative data	Primary control
External	Articles, advisories, feeds	Source allowlist + license
Enterprise	BOM, inventory, orders, routes	Role/field access
Reasoning	Events, graph, scenarios	Evidence + confidence
Action	Alerts, drafts, tasks	Approval + audit

TABLE III. DATA LAYERS AND CONTROLS

D. Supply-Chain Knowledge Graph

The graph nodes represent Supplier, Site, Part, Material, Product, Route, Port, Inventory, Production Line, Customer order, Event, and Source. Edges encode relationships such as SUPPLIES, PRODUCED_AT, REQUIRES, SHIPS_VIA, STORED_AT, FULFILLS, LOCATED_IN, OWNED_BY, and AFFECTS. For example, if two nodes are product and route, then the relationship is SHIPS_VIA. Temporal validity permits the system to distinguish a current supplier from an expired contract. Evidence and confidence are attached to every externally inferred relationship.

Graph traversal identifies exposure paths that a flat alert cannot: *earthquake* → *industrial park* → *wafer fab* → *chip family* → *assembly site* → *finished product* → *customer order*. Knowledge-graph reasoning has been studied for discovering underlying supply-chain risks [6], while LLM-based extraction can extend mapping from public information [7]. Chainsilience AI combines those methods with private, authoritative enterprise edges.

E. Digital Twin

The knowledge graph first establishes the enterprise entities and their dependencies. Each operational entity in the graph such as a supplier, inventory, or route, is linked to a corresponding digital twin containing its current and historical operational state. Together, these connected entity twins form the enterprise supply-chain digital twin. It is a time-indexed computational representation of the operational network, not just a dashboard. It combines topology with state variables: on-hand and in-transit inventory, demand, yields, capacities, lead times, route schedules, recovery assumptions, and contractual priorities. Digital supply-chain twins support disruption experiments for risk analysis and mitigation simulations [4].

Entity/state	Example fields
Supplier-site	capacity, recovery curve, qualification
Part/material	BOM usage, substitutes, yield loss
Inventory	on-hand, committed, in-transit, expiry
Route	mode, lane, transit, cost, chokepoints
Order	quantity, due date, priority, penalty

TABLE IV. MINIMUM DIGITAL-TWIN STATE

Twin updates are event-sourced: each change records its source, effective time, ingestion time, and previous value. Scenario runs operate on versioned snapshots so that analysts can reproduce why a recommendation was made.

F. Simulation and Optimization

A discrete-time simulator propagates capacity loss, lead-time shifts, yield changes, transit delays, and allocation rules through the network. Candidate mitigations may include switching suppliers, rerouting, reallocating scarce inventory, rescheduling production, qualifying a substitute, splitting volume, or negotiating due dates. Optimization is constrained by approved suppliers, technical compatibility, minimum order quantities, capacity, cost ceilings, and customer priority.

G. Agent Orchestration

Specialized AI agents have narrow roles: **Scout** (Non-AI) retrieves potential events; **Extractor** structures claims; **Verifier** filters out faulty information; **Mapper** resolves graph entities; **Analyst** requests deterministic scenarios; **Planner** ranks feasible responses; and **Communicator** drafts internal alerts or supplier/customer emails. A policy gateway mediates all tools.

Model Context Protocol (MCP) is one possible interoperability layer for connecting AI applications to external resources, prompts, and tools [17]. In this design, an MCP server may expose a read-only inventory query or a draft-email operation. MCP is not itself a security boundary; authentication, authorization, validation, and approval remain application responsibilities.

Agent	Permitted output	Prohibited by default
Scout	Candidate URLs/events	Bypass source policy
Extractor	Schema + evidence spans	Invent missing fields
Analyst	Scenario request/result	Override calculator
Planner	Candidate mitigation actions	Commit purchase/order
Communicator	Draft message	External send

TABLE V. SEPARATION OF AGENT DUTIES

H. Explainability and Interface

Every alert presents six linked cards: event facts; evidence and source quality; affected graph path; operational assumptions; quantitative impacts; and response options. Confidence is decomposed into source reliability, extraction confidence, entity-match confidence, and model uncertainty. The interface avoids a single opaque 'AI confidence' number.

Each LLM recommended candidate action includes the expected service benefit, implementation cost, recovery time, net financial impacts, and reduced risk. Users may change assumptions and rerun the scenario. A comparison view shows the baseline and each mitigation strategy under the same snapshot.

I. Alert and Communication Flow

Alerts are severity- and role-aware. Low-confidence events request verification, material events notify the designated incident channel, and external communications remain strictly reviewable drafts until approval. The system leverages Natural Language Processing (NLP) to dynamically generate context-aware email drafts that populate key operational details for external stakeholders**.** For suppliers, NLP constructs targeted inquiries requesting facility status, available capacity, revised lead times, and the next update window. For customers, NLP generates professional status updates that disclose only authorized facts, tailor information to the specific order context, and strictly avoid unsupported certainty.

The design converts AI-generated output into a reviewable draft rather than an irreversible action. Each approval records the reviewer, timestamp, message version, recipients, attached evidence, and downstream tool result. Human approval supports accountable oversight and reduces unauthorized action risk but does not by itself guarantee ethical or correct outcomes. NLP and LLM components strictly provide decision support.

IV. RISK ANALYSIS AND RESPONSE GENERATION

A. Risk Decomposition

Risk is modeled as a set of decision variables, not a free-form LLM judgment. For event *e* and asset, part, or order *i*, the normalized score is:

$$R_{i,e} = 100 \times C_e \times \left[w_H H_e + w_X X_{i,e} + w_V V_i + w_T T_{i,e} - w_M M_{i,e}\right]$$

Here, C is evidence confidence; H is hazard severity and duration; X is exposure through location, route, or dependency; V is vulnerability from concentration, substitutability, and recovery time; T is time criticality from demand and inventory; and M is mitigation strength. Inputs are scaled to [0,1], w refers to respective weights for each risk factor, which are versioned by use case, and the bracketed term is a weighted sum clipped to [0,1]. The score ranks attention. The LLM extracts evidence-backed risk metrics and company-specific factors such as expected delay duration, revenue at risk, inventory coverage, and the absence of qualified alternative suppliers, and structures them as inputs for the deterministic risk engine (the equation above). After the score is calculated, the LLM reasons how different factors synthesized to produce the risk score and risk level.

Fig. 2. Configurable triage bands. Thresholds route work; they do not replace review.

B. Operational Metrics

Days of cover for part *p* is **DOC = available inventory / protected daily demand**. Available inventory excludes quarantined stock and can apply yield, expiry, and allocation factors. Projected shortage date is the first period where cumulative qualified supply falls below cumulative protected demand. Expected incremental cost for an action is **EIC = premium purchase + logistics + qualification + changeover + penalties - avoided loss**. Values are displayed as ranges where costs or recovery times are uncertain. Financial assumptions are stored separately from source facts.

C. Production Stoppage Probability

Production stoppage probability is estimated by a probabilistic simulation service using Monte Carlo sampling over distributions for disruption duration, recovery time, transit delay, yield, demand, and alternative capacity. The Analyst agent supplies validated scenario parameters, invokes the simulation service, and explains the results; the LLM does not perform the numerical simulation itself. The displayed value is the fraction of runs in which a protected line cannot meet its minimum schedule. Calibration is assessed on historical or synthetic labeled incidents; absent such evidence, the interface must label the output a scenario frequency rather than an empirical probability.

D. Response Generation

The Planner receives only verified facts, twin state, scenario results, and constraints. It generates a diverse set of candidate mitigation options, and invokes a deterministic checker which ranks the options by a multi-objective score:

$$\begin{aligned} U(a) = {} & \beta S \cdot \mathrm{Service}(a) + \beta F \cdot \mathrm{NetFinancialImpact}(a) \\ & + \beta R \cdot \mathrm{ReducedRisk}(a) - \beta I \cdot \mathrm{ImplementationCost}(a) \end{aligned}$$

The interface shows each component instead of hiding the weighting. A user can select service-first, cost-first, or balanced policy profiles. No alternative supplier is recommended solely because it appears in public text: the supplier must satisfy enterprise approval, technical qualification, sanctions/export-control checks, capacity, commercial, and cybersecurity requirements.

Action class	Feasibility checks	Approval
Expedite/reroute	lane, carrier, customs, cost	Logistics owner
Reallocate inventory	priority, contract, service	Operations
Switch supplier	quality, capacity, legal, IP	Cross-functional
Reschedule	line, tooling, labor, due dates	Plant planner
Customer notice	facts, disclosure, commitment	Account owner

TABLE VI. ACTION CONTROLS

E. RAG-Grounded Mitigation Planning

During mitigation planning, RAG retrieves authorized, current records—including approved suppliers, qualified substitute components, contracts, company policies, available capacities, logistics routes, and previous disruption responses—and provides them as evidence to the LLM-based Planner [9]. The LLM uses this retrieved context to reason about and propose candidate mitigation actions, while deterministic feasibility checks and optimization services validate constraints and compare the candidate options based on service recovery, implementation cost, net financial impacts, and reduced risk. Each recommendation retains links to the retrieved records so decision-makers can verify why the action was proposed; missing, stale, or conflicting records trigger review rather than unsupported recommendations.

F. Business Value

The value hypothesis has four components: reduced time-to-awareness; reduced time-to-impact assessment; lower expected disruption cost through earlier optionality; and more consistent, auditable communication. Benefits should be measured against a baseline workflow rather than inferred from model novelty.

A participating company provides authorized supply-chain data, operational constraints, and decision rules. Chainsilience AI converts these inputs into earlier warnings, quantified exposure, and feasible mitigation options, while the company retains decision authority and supplies outcome data that improves future scenario calibration. Value is measured through avoided expected disruption loss, analyst hours saved, and improved service continuity relative to implementation and operating cost.

For an SME, a useful deployment can begin with one product family, a limited supplier set, inventory and purchase-order extracts, and a small number of trusted external sources. This minimizes integration cost and gives the team a measurable pilot. Semiconductor deployments can then extend to multi-tier material, utility, equipment, foundry, assembly/test, and route dependencies.

G. Alert Content

A complete alert states: **signal** (what changed); **exposure** (the graph path); **impact** (orders, dates, cost, uncertainty); **options** (ranked and constrained); **decision** (owner and deadline); and **evidence** (sources and retrieved passages). The same structured object powers the dashboard, internal notification, and email draft, preventing channel-to-channel inconsistency.

Where sources disagree, the alert shows the conflict and requests confirmation. Where internal data are stale, it displays the last refresh time. Where no feasible mitigation exists, it recommends escalation rather than fabricating an alternative.

V. CASE STUDY AND EVALUATION

A. Evidence-Based Trigger

On 3 April 2024, a magnitude 7.4 earthquake occurred near Hualien City, Taiwan, followed by strong aftershocks [11]. TSMC later reported that earthquakes in April 2024 caused damage to inventories, facilities, machinery, and equipment and that the company recognized approximately NT$3 billion in losses, net of insurance claims, in the second quarter of 2024 [12]. These facts establish a real disruption context, but they do not establish the hypothetical buyer impacts below.

B. Illustrative Enterprise Scenario

Assume a fictional electronics SME, *Orion Controls*, buys a controller chip through Supplier A located in Taiwan. The enterprise twin states that Supplier A represents 42% of qualified wafer volume; usable inventory is 12 days; protected demand is 8,000 units/day; Supplier B is approved for 25% incremental volume after a five-day changeover; and premium logistics plus changeover would cost USD 180,000. These values are design-test inputs, not claims about any real company.

Stage	Illustrative output
Verify	M7.4 event; location/time confirmed; confidence 0.94
Map (Knowledge Graph)	Event -> Taiwan site -> Supplier A -> chips -> 3 SKUs -> 7 orders
Simulate	Baseline shortage day 13; 3-week recovery range
Estimate	71% chance of production stoppage within 21 days
Plan	Shift 25% to Supplier B; expedite 6 days; reallocate stock
Cost	USD 180k estimate; range USD 150k-230k

TABLE VII. ILLUSTRATIVE EVENT-TO-ACTION TRACE

The system would display the evidence supporting the earthquake, the enterprise records supporting the dependency and inventory figures, the simulation version, and the scenario frequency by Monte Carlo simulation producing the 71% chance of production stoppage. A user can compare 'wait,' 'expedite,' 'reallocate,' and 'switch 25%' alternatives before approving supplier outreach.

C. Example Communication

The Communicator drafts a supplier-status request that cites the affected part and asks for facility status, available capacity, recovery process, allocation policy, and next update time. It does not state that a three-week delay is confirmed. The procurement owner edits and approves the draft; the system records the sent version and any reply as new evidence.

D. Evaluation Design

Evaluation uses a replay corpus of dated disruption events and frozen enterprise snapshots. Articles arriving after each decision cutoff are withheld to prevent future information leakage. Two analysts independently label event fields and relevant graph paths; disagreements are adjudicated. Synthetic events supplement rare categories but are reported separately.

Dimension	Metric	Pilot target*
Detection	Detecting disruptions 24h; false alerts/day	>=85%; <=2
Extraction	Correct extraction of Entity/relation rate	>=85%
Verification	Unsupported-claim rate	<2%
Mapping	Disruption connected to correct supply chain path rate	>=80%
Risk ranking	Precision@K; rank correlation	Improve baseline
Production stoppage probability	Brier score; calibration error	Improve baseline
Impact	Accuracy for delay/DOC	Use-case set
Action	Feasible recommendation rate	>=90%
Workflow	Median alert-to-decision time	-50% vs baseline
Safety	Unauthorized action rate	0

TABLE VIII. PROPOSED PILOT METRICS. *Targets are acceptance criteria, not achieved results.

Baselines are: keyword alerts plus manual analysis; LLM extraction without verification; and risk ranking without a digital twin. Component-removal tests remove provenance, graph context, or deterministic simulation to quantify their contribution. Analysts blind-score relevance, traceability, feasibility, and clarity. Operational users complete timed scenario tasks and a structured usefulness survey.

E. Feasibility and Implementation

A 12- to 16-week pilot is proposed. Weeks 1-3 define the ontology, data-sharing boundary, and success metrics. Weeks 4-7 integrate a narrow supplier/product scope and build event extraction. Weeks 8-10 add graph mapping and deterministic inventory/lead-time calculations. Weeks 11-13 add scenario comparison, dashboard, and draft communications. Weeks 14-16 run replay tests, red-team exercises, and user acceptance.

The technical stack can use a relational store for enterprise records, a graph SQL database for dependencies, object storage for source documents, a queue for event processing, a simulation service, and a web dashboard. Model choice remains replaceable. Deployment may be cloud (via AWS), virtual private cloud, or on-premises depending on confidentiality and integration constraints.

F. Evaluation Threats

Historical coverage may favor highly reported events; ground truth for supplier impact may be unavailable; analysts may disagree on relevance; and synthetic scenarios may not match real behavior. Results therefore require confidence intervals, category-level reporting, dataset documentation, and clear separation between retrospective accuracy and prospective operational value.

VI. GOVERNANCE, LIMITATIONS, AND CONCLUSION

A. Data, Security, and Privacy

The governance design follows the NIST AI RMF functions - Govern, Map, Measure, and Manage [13] - and addresses generative-AI risks such as confabulation and information integrity [14]. Cybersecurity controls map to NIST CSF 2.0 [15]. Zero-trust principles require explicit, continuously evaluated access to resources rather than trust based on network location [16].

Controls include source allowlists; data minimization; encryption; tenant isolation; role- and attribute-based access; secret management; software and model inventories; prompt/input sanitization; network egress restrictions; immutable audit events; retention schedules; backups; incident response; and periodic penetration and dependency testing. Sensitive supplier prices, forecasts, and customer orders are excluded from prompts unless strictly required and contractually permitted.

B. Hallucination and Source Controls

Generated facts must cite retrieved evidence. Structured fields are schema-validated; numeric values are recalculated outside the LLM; dates and units are normalized; contradictions trigger review; and low-confidence mappings cannot automatically propagate to high-severity actions. The interface distinguishes quoted source facts, enterprise facts, model inferences, scenario assumptions, and user decisions.

C. Human Oversight and Ethics

Consequential decisions remain accountable to named roles. Procurement commitments, supplier changes, customer promises, and external emails require approval. Recommendations should be checked for systematic disadvantage to small or geographically concentrated suppliers, and users must not treat public reporting as proof of wrongdoing. Source licensing, privacy, export-control, sanctions, competition, and contractual duties require jurisdiction-specific review. Chainsilience AI must securely protect sensitive company data, including suppliers, routes, products, orders, inventory, and production schedules, using encryption and strict access controls. This information must not be disclosed to third parties or used to train external AI models without explicit company consent.

D. Limitations

Chainsilience AI cannot observe undisclosed multi-tier dependencies, guarantee source truth, predict unprecedented events with precision, or make unqualified suppliers immediately usable. Digital twins decay when master data are stale. LLM outputs remain probabilistic; simulation outputs inherit uncertain distributions; and optimizing one firm's resilience can shift cost or scarcity to others. The system is decision support, not autonomous authority.

Acknowledgment- Prepared as a project white paper for IDSOL 2026.

E. Future Work

Future work includes using MCPs to connect LLM to email sending services such that an agent can autonomously send customer/supplier email after human approval, probabilistic entity resolution, richer multi-tier mapping, causal recovery models, federated or privacy-preserving learning, active learning from analyst corrections, robust optimization, multilingual evaluation, and standardized connectors. A mature deployment should measure not only detection quality but whether decisions reduce expected loss without creating unacceptable security, fairness, or compliance risk.

F. Conclusion

Chainsilience AI proposes a practical bridge between disruption news and operational response. Its architecture combines verified event extraction, provenance, a supply-chain knowledge graph, a digital twin, deterministic and probabilistic analytics, constrained planning, and approval-gated agentic communication. The design is deliberately hybrid: LLMs interpret language and draft explanations; graphs preserve relationships; simulators estimate consequences; solvers test feasible mitigations; and humans retain authority.

The central test for IDSOL 2026 is therefore not whether the system can produce a persuasive alert. It is whether that alert is earlier than the baseline, correctly linked to the company, quantitatively reproducible, operationally useful, secure, and safe to act upon. The proposed replay study and staged pilot make those claims falsifiable.

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ANNEX 1

CHAINSILIENCE AI

Glossary and Technical Implementation Annex

A governed agentic, knowledge-graph, and digital-twin framework for supply-chain disruption intelligence

Submitted as part of the project white paper for the International Data Science Olympiad (IDSOL) 2026

Ka Woon Ho | Ke Xin Cao | Gary Leng | Mo Xia | Andrew Li

Aligned with Whitepaper

Purpose of this annex
This annex defines the white paper's keywords, expands the proposed system workflow, and records the data, analytical, governance, and pilot details needed to interpret the architecture. It does not convert illustrative values or pilot targets into achieved results.

1. Annex Scope and Reading Guide

The main white paper presents the competition narrative and six-page architecture. This annex provides the supporting definitions and implementation detail. Its terminology and status labels follow the most recent revised white paper.

1.1 Status labels used throughout

Label	Meaning in this annex
Implemented / prototyped	A module or interface demonstrated by the current functional prototype, limited to what actually exists.
Planned	An architecture component, enterprise integration, simulation, scoring method, or controlled action proposed for later implementation.
Assumed	Required enterprise data, permissions, constraints, or scenario inputs supplied for design and evaluation.
Illustrative	A case-study value used to demonstrate workflow behavior; it is not a measured result or a claim about a real company.

1.2 Workflow at a glance

1. Detect and verify	2. Map exposure	3. Analyze impact	4. Plan and approve
Collect permitted sources; filter relevance; extract and cross-check claims.	Resolve entities; traverse Supplier -> Part -> Product -> Order dependencies.	Update the digital twin; calculate inventory, delays, cost, and scenario risk.	Retrieve approved options; test feasibility; compare mitigations; obtain authorization.

1.3 Core design boundary

Hybrid decision-support architecture

LLMs interpret language, retrieve and organize evidence, propose candidate actions, and draft explanations. Databases preserve records; the graph stores relationships; deterministic services calculate quantities; simulation estimates scenario outcomes; optimization compares feasible actions; policy controls permissions; and named humans retain decision authority.

2. Concise Keyword Dictionary

The definitions below correspond exactly to the ten keywords listed near the start of the revised white paper. Each entry states what the term means and why it matters in Chainsilience AI.

- **Supply-chain resilience:** The ability of a supply network to anticipate, absorb, adapt to, and recover from disruption while maintaining critical service.
  - **Use and significance:** It is the system's intended outcome. Chainsilience AI improves resilience by shortening detection and assessment time, quantifying exposure, and comparing feasible mitigations before operations stop.
- **Agentic AI:** AI organized into goal-directed agents that reason over evidence and use permitted tools to complete bounded tasks.
  - **Use and significance: A non-AI Scout agent polls approved RSS news feeds; the Extractor, Verifier, Mapper, Analyst, Planner, and Communicator are AI agents that divide the remaining workflow into auditable duties. Narrow permissions and human approval reduce the risk of unrestricted autonomous action.**
- **Digital twin:** A time-indexed computational representation of the current supply-chain network and its operating state.
  - **Use and significance:** The twin combines inventory, demand, orders, capacity, yield, lead times, routes, and recovery assumptions. It lets the system replay disruptions and compare no-action and mitigated scenarios on a reproducible snapshot.
- **Knowledge graph:** A graph of supply-chain entities and the relationships between them, with evidence, confidence, and temporal validity.

  - **Use and significance:** It connects an external event to affected suppliers, sites, parts, products, routes, production lines, and customer orders. This converts a general news alert into a company-specific exposure path.
- **Semiconductor risk:** Operational risk arising from concentrated fabrication and assembly capacity, specialized materials and equipment, long qualification cycles, geopolitical controls, and limited substitutability.
  - **Use and significance:** Chainsilience AI represents these dependencies and constraints so that alternatives are not recommended merely because they appear in public information.
- **Retrieval-augmented generation (RAG):** A method that retrieves relevant, current records and supplies them to an LLM before it generates an output.
  - **Use and significance:** RAG grounds event summaries, exposure explanations, and mitigation proposals in source-linked news and authorized enterprise records. During planning it retrieves approved suppliers, qualified substitutes, contracts, policies, capacity, routes, and previous responses.
- **Large language model (LLM) reasoning:** The use of an LLM to interpret unstructured language, relate retrieved evidence, propose options, and explain conclusions.
  - **Use and significance:** The LLM filters relevance, extracts events, assists verification and mapping, reasons over retrieved constraints, and drafts alerts or emails. It does not calculate inventory, run Monte Carlo simulation, solve optimization, or approve consequential actions.
- **Small and medium-sized enterprise (SME):** A business with fewer financial, analytical, and operational resources than a large enterprise, with thresholds varying by jurisdiction.
  - **Use and significance:** SMEs are a primary target because many rely on supplier emails, spreadsheets, search alerts, and periodic meetings. A narrow product-and-supplier pilot can provide earlier analysis without requiring a large risk-intelligence team.
- **Model Context Protocol (MCP):** An interoperability protocol through which an AI application can access defined resources, prompts, and tools through structured interfaces.
  - **Use and significance: In the current prototype, MCP may support read-only queries or draft-email creation; emails remain reviewable drafts. A future MCP connector may send the exact approved draft only after human review and approval. Authentication, authorization, validation, least privilege, and approval remain the application's responsibility.**
- **Natural language processing (NLP):** Methods that allow computers to analyze, structure, and generate human language.
  - **Use and significance:** NLP supports language detection, relevance classification, entity and relation extraction, summaries, alerts, and context-aware supplier or customer email drafts. It turns fragmented text into structured, reviewable information.

3. Detailed End-to-End Execution Workflow

The following sequence expands Section II.D and Section III of the white paper. Every transition records provenance, confidence, timestamp, user or agent identity, and version where applicable.

Step 1 - Establish authority and scope

A company administrator identifies data owners and decision approvers, then defines the permitted systems, business units, products, suppliers, locations, fields, users, retention periods, and approval thresholds. Connections operate with least-privilege access. The system records purpose, consent, access policy, and data lineage.

Step 2 - Upload or synchronize authorized company data

The company supplies relevant products, parts, bills of materials, supplier sites, qualified alternatives, inventory, in-transit stock, purchase and customer orders, production schedules, capacities, yields, logistics routes, lead times, costs, contracts, and service priorities. A pilot may begin with secure CSV templates; later versions may use read-only enterprise integrations.

Step 3 - Validate and normalize the data

The ingestion layer standardizes identifiers, units, currencies, dates, and time zones and detects duplicates, missing links, stale balances, impossible values, and contradictory supplier records. Significant quality problems are quarantined or presented to an authorized user. Unverified records do not silently influence high-impact recommendations.

Step 4 - Build the operational baseline

Validated records form a company-specific knowledge graph and a versioned digital-twin snapshot. Users inspect supplier-to-order paths, confirm alternatives and routes, check data freshness, and preserve the precise state used for later calculations.

Step 5 - Detect and verify an external disruption

The non-AI Scout agent polls approved RSS news feeds, de-duplicates entries, and stores source metadata; it does not judge relevance or verify claims. An LLM-based agent then filters company relevance and extracts structured event fields. A Verifier compares independent sources, checks temporal and geographic consistency, identifies contradictions, and flags unsupported or low-confidence claims for review.

Step 6 - Map exposure and calculate the no-action impact

The verified event is resolved against internal suppliers, sites, routes, ownership, parts, products, facilities, and customer commitments. Graph traversal identifies affected paths. Deterministic services and probabilistic simulation estimate inventory coverage, shortage and delay windows, production interruption scenarios, service exposure, incremental cost, and uncertainty if the company takes no action.

Step 7 - Generate and compare mitigation strategies

RAG retrieves current approved suppliers, qualified substitutes, contracts, policies, capacities, routes, and previous responses. The LLM-based Planner proposes candidate actions. A deterministic feasibility checker rejects options that violate qualification, capacity, commercial, legal, export-control, sanctions, budget, cybersecurity, or approval constraints. Simulation and optimization compare the remaining options.

Step 8 - Review, revise, and approve

A named decision-maker reviews sources, affected paths, enterprise records, freshness, assumptions, uncertainty, calculations, and alternatives. The user may correct data, change assumptions, edit an action, select another option, or rerun the scenario. Approval records the exact scope, budget, recipients, message version, and decision owner.

Step 9 - Execute controlled actions

The current implementation releases only internal, policy-permitted outputs after approval, such as creating procurement or planning tasks, notifying an internal team, or saving an approved supplier-status email as a reviewable draft. Chainsilience AI does not send external emails. In a future version, an MCP connector may transmit the exact approved draft only after an authorized reviewer approves its content, recipients, and scope. Every action, owner, deadline, response, confirmation, failure, and status change is recorded.

Step 10 - Observe outcomes and learn

The dashboard distinguishes the no-action baseline, the projected mitigated outcome, and the realized outcome. Supplier confirmations, inventory changes, shipment events, production records, receipts, and order data update the twin. Projected benefit is not reported as realized improvement until operational evidence supports it.

4. DSAI Components and Responsibility Boundaries

Chainsilience AI uses a hybrid DSAI architecture. The following map prevents the LLM from being credited with calculations or authority that belong to other components.

Component	Primary use in Chainsilience AI	Boundary / significance
Non-AI Scout and connectors	Poll approved RSS news feeds and retrieve minimum required internal fields.	The Scout does not classify, reason, verify, or generate; LLMs do not scrape or authenticate data sources.

Component	Primary use in Chainsilience AI	Boundary / significance
NLP and LLM agents	Filter relevance, extract event facts, compare claims, assist entity mapping, propose options, explain results, and draft communications.	Outputs remain probabilistic and evidence-linked; consequential actions require controls and approval.
RAG	Retrieve verified evidence and authorized company records before summaries, explanations, or mitigation proposals are generated.	Reduces unsupported output and makes recommendations traceable to current records.
Knowledge graph	Store multi-tier dependency relationships, provenance, confidence, and temporal validity.	Supports explainable company-specific exposure paths rather than flat keyword alerts.
Digital twin	Represent the operational state at a known time and provide reproducible scenario snapshots.	Separates current state, scenario assumptions, and later realized observations.
Deterministic calculations	Calculate days of cover, shortage dates, lead-time effects, costs, and constraint checks.	Numbers are recalculated outside the LLM.
Probabilistic simulation	Estimate scenario frequency and uncertainty through Monte Carlo sampling.	The Analyst agent supplies validated parameters and explains results; it does not perform the numerical simulation.
Optimization service	Compare feasible actions across service, cost, residual risk, and implementation time.	Only qualified and policy-compliant candidates enter the solver.
Policy gateway and approval	Enforce permissions, action limits, recipients, budgets, and human authorization.	MCP or agent orchestration is not itself a security boundary.

4.1 Component and AI-agent separation of duties

Component / agent	Permitted responsibility	Prohibited by default
Scout (non-AI)	Poll approved RSS news feeds; de-duplicate and store candidate items with source metadata.	Infer relevance, verify claims, or access sources outside the approved RSS allowlist.
Extractor	Produce a structured schema with cited evidence spans.	Fill missing fields with unsupported content.
Verifier	Compare sources and flag unreliable, contradictory, unsupported, or low-confidence claims.	Guarantee that a report is true or false without adequate corroboration.
Mapper	Resolve event entities against the authorized company graph.	Create high-impact graph facts from an unresolved match.
Analyst	Request deterministic and probabilistic scenarios and interpret returned results.	Override calculators or simulate numerically inside the LLM.
Planner	Propose evidence-grounded candidate mitigations.	Rank or select candidates using quantitative impact metrics, commit purchases, change suppliers, or bypass feasibility checks.
Communicator	Draft internal alerts and supplier or customer messages.	Send or dispatch external emails; outputs remain reviewable drafts.

4.2 RAG-grounded mitigation planning

During mitigation planning, RAG retrieves only authorized and current records relevant to the verified disruption. These records may include approved suppliers, qualified substitute components, contracts, company policies, available capacities, logistics routes, customer priorities, and previous disruption responses. The LLM-based Planner reasons over this retrieved context to propose candidate actions. Deterministic feasibility checks, simulation, and optimization then validate constraints and compare service recovery, cost, implementation time, and residual risk. Missing, stale, or conflicting records trigger review rather than an unsupported recommendation.

5. Data, Graph, and Digital-Twin Specification

5.1 Input data boundary

Layer	Representative fields	Primary control
External evidence	Articles, government advisories, earthquake and weather feeds, trade notices, port and carrier updates, company disclosures, licensed datasets.	Source allowlist, licensing, hashing, timestamping, language detection, deduplication, provenance.
Enterprise master data	Suppliers, sites, products, parts, materials, bills of materials, ownership, approved alternatives.	Role- and field-level access; validation; change history.
Operational state	Inventory, in-transit stock, orders, schedules, capacities, yields, routes, lead times, costs, priorities.	Freshness checks, snapshot versioning, purpose limitation.
Reasoning artifacts	Events, entity matches, graph paths, assumptions, scenario inputs, results, recommendations.	Evidence links, confidence decomposition, schema validation, version control.
Action records	Approvals, drafts, recipients, tasks, tool results, confirmations, failures.	Policy enforcement, immutability, audit, retention.

5.2 Event extraction schema

A relevant external item is converted into a structured event object containing: event type; source and evidence span; affected organization, facility, location, route, commodity, or product; start and observation time; expected duration or recovery range; severity indicators; reported quantities; source reliability; extraction confidence; contradiction status; and review state. Unknown values remain unknown rather than being completed by the LLM.

5.3 Knowledge-graph model

Graph element	Examples	Why it matters
Entities	Supplier, Site, Part, Material, Product, Route, Port, Carrier, Inventory Node, Production Line, Order, Customer, Event, Source.	Provides the objects needed to connect a disruption to operations.
Relationships	SUPPLIES, PRODUCED_AT, REQUIRES, SHIPS_VIA, STORED_AT, FULFILLS, LOCATED_IN, OWNED_BY, AFFECTS.	Represents multi-tier dependency paths that a flat database alert may hide.
Evidence and confidence	Source record, cited passage, extraction confidence, entity-match confidence, reviewer confirmation.	Allows every externally inferred relationship to be challenged and traced.
Temporal validity	Effective-from, effective-to, observation time, ingestion time.	Distinguishes a current supplier or route from an expired relationship.

Example exposure path

Earthquake -> Taiwan industrial area -> Supplier A site -> controller chip -> three products -> seven customer orders. The path explains why a geographically severe event is operationally relevant to this specific company.

5.4 Minimum digital-twin state

Entity or state	Minimum scenario fields
Supplier and site	Capacity, recovery curve, qualification status, concentration, ownership.
Part and material	BOM usage, compatibility, substitute status, yield loss, qualification time.
Inventory	On-hand, committed, quarantined, in-transit, expiry, allocation.
Route	Mode, lane, transit time, cost, schedule, chokepoints, customs constraints.
Production	Line, schedule, minimum run, yield, tooling, labor, alternate capacity.
Order and customer	Quantity, due date, priority, service commitment, penalty.

Twin updates are event-sourced. Each change records its source, effective time, ingestion time, previous value, and authorizing identity. Scenario runs use frozen versions so users can reproduce why a recommendation was made.

6. Quantitative Risk and Mitigation Logic

6.1 Risk decomposition

The normalized triage score ranks attention; it is not a probability. For event e and affected item i, the white paper combines evidence confidence (C), hazard severity and duration (H), exposure (X), vulnerability (V), time criticality (T), and mitigation strength (M) using versioned weights. Inputs are scaled to [0, 1], and the weighted result is clipped to [0, 1].

Interpretation rule

Risk rank answers 'what should be reviewed first?' Production-stoppage probability answers 'how often does a protected line fail under the stated scenario distributions?' These outputs use different metrics and must not be conflated.

6.2 Operational calculations

Measure	Definition	Decision use
Days of cover (DOC)	Available qualified inventory divided by protected daily demand, adjusted for quarantine, yield, expiry, and allocation.	Shows how long operations can continue before replenishment is required.
Projected shortage date	First period in which cumulative qualified supply is lower than cumulative protected demand.	Identifies the decision deadline and affected orders.
Expected incremental cost (EIC)	Premium purchase + logistics + qualification + changeover + penalties - avoided loss.	Compares mitigation expense with expected disruption loss.
Stoppage scenario frequency	Fraction of Monte Carlo runs in which a protected line cannot meet its minimum schedule.	Quantifies uncertainty under explicit duration, transit, yield, demand, and alternative-capacity distributions.
Action utility	Service benefit - cost - residual risk - implementation time, using visible policy weights.	Supports service-first, cost-first, or balanced comparison without hiding trade-offs.

6.3 Candidate action classes and controls

Action class	Required feasibility checks	Approval owner
Expedite or reroute	Lane, carrier, customs, capacity, cost, schedule.	Logistics owner
Reallocate inventory	Customer priority, contract, protected demand, service impact.	Operations owner
Switch or split supplier	Technical qualification, capacity, legal, sanctions, export control, IP, cybersecurity.	Cross-functional authority
Reschedule production	Line, tooling, labor, yield, due dates, changeover.	Plant planner
Customer or supplier notice	Verified facts, disclosure limits, commitments, recipients, approved wording.	Account or procurement owner

6.4 Decision-state reporting

Displayed state	Meaning
No-action baseline	Estimated impact if no mitigation is implemented under the selected snapshot and assumptions.
Projected mitigated outcome	Expected result of the approved strategy before sufficient operational evidence has arrived.
Realized outcome	Observed result supported by confirmations, inventory, shipment, production, receipt, and order records.

7. Explainability, Security, and Human Oversight

7.1 Evidence chain presented to the user

Interface card	Required content
Event facts	What happened, where, when, reported duration, and unresolved fields.
Evidence and source quality	Retrieved sources, cited passages, contradictions, and reliability indicators.
Affected graph path	The supplier, site, route, part, product, production, and order links that create exposure.
Operational assumptions	Data freshness, scenario inputs, distributions, user edits, and model versions.
Quantitative impact	Inventory coverage, shortage dates, delays, cost ranges, scenario frequency, and uncertainty.
Response options	Benefits, costs, implementation time, prerequisites, reversibility, owner, residual risk, and approval state.

7.2 Security and confidentiality controls

Sensitive company data - including suppliers, routes, products, orders, inventory, production schedules, prices, forecasts, and customer commitments - remains within the protected company environment and is used only for authorized supply-chain analysis. Controls include encryption, tenant isolation, role- and attribute-based access, secret management, source allowlists, data minimization, prompt and input sanitization, network egress restrictions, immutable audit events, retention schedules, backups, incident response, and periodic security testing. Internal data must not be disclosed to third parties or used to train external AI models without explicit company consent.

7.3 Approval and audit requirements

In the current implementation, external emails remain editable drafts even after a reviewer approves their content; Chainsilience AI does not send them. Procurement commitments, supplier changes, and customer promises require approval by named roles. A future MCP sending capability may transmit only the exact approved draft after explicit review of its recipients, content, and scope. The approval record includes the reviewer, time, exact action or message version, recipients, budget, evidence, and downstream result. Expired, changed, or previously consumed approvals are rejected.

7.4 Failure and escalation behavior

- **Low-confidence event:** Request verification instead of generating a definitive high-severity warning.
- **Unresolved entity:** Place the match in a review queue; do not silently create a graph relationship.
- **Stale internal data:** Display the last refresh time and block or qualify calculations that depend on freshness.
- **Conflicting sources:** Show the conflict, retain both evidence paths, and request confirmation.
- **No feasible mitigation:** Escalate the constraint conflict instead of fabricating an alternative.
- **Tool or workflow failure:** Record the failure, preserve idempotency, notify the responsible owner, and avoid partial unauthorized action.

8. Pilot Evaluation and Implementation Plan

8.1 Meaning of the pilot

The pilot is a small, controlled validation project before company-wide deployment. It may cover one company or business unit, one product family, a limited supplier and route set, selected disruption categories, a fixed historical replay period, and a small group of operational users. The targets below are acceptance criteria, not achieved results.

8.2 Evaluation protocol

- Replay dated historical disruption events against frozen enterprise snapshots.
- Withhold articles published after each decision cutoff to prevent future-information leakage.

- Have two analysts independently label event fields and affected graph paths; adjudicate disagreements.
- Report synthetic rare-event scenarios separately from historical-event performance.
- Compare against keyword alerts plus manual analysis, LLM extraction without verification, and risk ranking without a digital twin.
- Use component-removal tests to measure the contribution of provenance, graph context, and deterministic simulation.
- Have analysts blind-score relevance, traceability, feasibility, and clarity; have operational users complete timed scenario tasks and a structured usefulness survey.

8.3 Proposed pilot metrics

Dimension	Metric	Pilot acceptance target
Detection	Recall within 24 hours; false alerts per day	>=85%; <=2
Extraction	Entity and relation F1	>=85%
Verification	Unsupported-claim rate	<2%
Mapping	Correct affected-path rate	>=80%
Risk ranking	Precision@K; rank correlation	Improve baseline
Production-stoppage probability	Brier score; calibration error	Improve baseline
Impact	Mean absolute error for delay and DOC	Threshold set by use case
Action	Feasible recommendation rate	>=90%
Workflow	Median alert-to-decision time	50% lower than baseline
Safety	Unauthorized action rate	0

Recall within 24 hours measures detection timeliness; F1 balances missed and incorrect extractions; Precision@K measures whether top-ranked risks are important; Brier score and calibration error assess probability quality; mean absolute error measures average prediction error; and DOC means days of cover.

8.4 Proposed 12- to 16-week pilot

Period	Primary work	Exit evidence
Weeks 1-3	Define ontology, data-sharing boundary, users, approval rules, baseline, and acceptance criteria.	Authorized scope, data dictionary, evaluation plan.
Weeks 4-7	Integrate a narrow supplier and product scope; build collection, relevance filtering, extraction, and verification.	Versioned events with cited evidence and quality checks.
Weeks 8-10	Add graph mapping and deterministic inventory and lead-time calculations.	Traceable affected paths and reproducible impact calculations.
Weeks 11-13	Add scenario comparison, RAG-grounded planning, dashboard, and reviewable communications.	Feasible options, approval flow, audit records.
Weeks 14-16	Run replay tests, red-team exercises, user acceptance, and metric review.	Pilot report with achieved results separated from targets.

9. Illustrative Taiwan-Earthquake Scenario

This scenario expands the white paper's retrospective example. The 3 April 2024 Taiwan earthquake is a real evidence-based trigger, while Orion Controls, its dependencies, probabilities, quantities, costs, and allocation choices are fictional design-test inputs.

9.1 Scenario assumptions

Input	Illustrative value
Protected part	Controller chip used by three fictional product SKUs.
Primary dependency	Supplier A in Taiwan provides 42% of qualified wafer volume.
Inventory	12 days of usable coverage at protected demand of 8,000 units per day.
Alternative	Supplier B is approved for 25% incremental volume after a five-day changeover.
Mitigation cost	USD 180,000, with an illustrative range of USD 150,000-230,000.
Scenario output	71% of simulation runs show production stoppage within 21 days under the stated distributions.

9.2 Event-to-decision trace

Stage	System behavior	Evidence retained
Collect	The non-AI Scout polls approved RSS feeds for permitted earthquake and company-impact reports.	Source URL, hash, publication and ingestion times.
Filter and verify	Classify company relevance; extract location, time, severity, and affected entities; cross-check sources.	Cited passages, contradiction state, confidence components.
Map	Resolve the event to Supplier A's site and traverse chip, product, and order dependencies.	Entity matches and Supplier -> Part -> Product -> Order path.
Simulate no action	Estimate shortage day, recovery range, affected orders, and stoppage scenario frequency.	Frozen twin version, distributions, calculation and simulation versions.
Plan	Retrieve approved alternatives and propose wait, expedite, reallocate, and switch-25% options.	Supplier qualification, capacity, route, contract, and policy records.
Compare	Test constraints and compare service recovery, cost, time, reversibility, and residual risk.	Feasibility results, solver inputs, ranked alternatives.
Approve and communicate	A decision-maker selects the exact option and approves a reviewable supplier-status draft.	Reviewer, action scope, recipients, message version, timestamp.
Observe	Update the twin with supplier replies, shipment status, inventory, production, and order outcomes.	Projected versus realized results and before-after audit trail.

9.3 Example communication boundary

Reviewable supplier-status request

The Communicator may draft questions about facility status, available capacity, recovery process, allocation policy, revised lead times, and the next update time. It must not state that a three-week delay is confirmed unless retrieved evidence supports that claim. The procurement owner may edit and approve the draft, but Chainsilience AI does not send it. A future MCP connector may send only the exact approved version after explicit review; any resulting reply can then be retained as new evidence.

10. Limitations and Annex Control Notes

Limitation	Required interpretation or control
Undisclosed multi-tier dependencies	The system cannot map relationships absent from public evidence and authorized enterprise data; unresolved exposure remains visible as uncertainty.
Source truth	Cross-source verification can reduce unsupported claims but cannot guarantee that every source is correct.
Stale enterprise records	Digital twins decay; freshness and version must accompany every scenario.
Unprecedented events	Historical and synthetic evaluation may not represent novel combinations of shocks.
Qualification constraints	A publicly mentioned supplier or component is not operationally usable until enterprise, technical, legal, commercial, and cybersecurity checks pass.
Model and simulation uncertainty	LLM outputs are probabilistic; simulations inherit uncertain distributions; results require ranges, sensitivity analysis, and human judgment.
System authority	Chainsilience AI is decision support, not autonomous procurement, production, or external-communication authority.

Reference note: bracketed sources, standards, and literature supporting these concepts are listed in the bibliography of the revised Chainsilience AI white paper. This annex intentionally avoids adding claims beyond that paper.